$$A = \begin{bmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & & & & & \\ 0 & 0 & \cdots & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & & & & & \\ \vdots & & & & & \\ 0 & - & - & - & - & - \end{bmatrix}_{n \times m}$$

(r th row)

$$= \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}_{n \times m}$$

$T : V \longrightarrow U$, $W \subseteq V$ 14.02.2020

Then restriction of T to W is the map

$T/W : W \longrightarrow U$ s.t $T/W(v) = T(v)$, for every $v \in W$

Prove that T/W is linear.

Sol:- (i) let $u_1, u_2 \in W$ s.t $T/W(u_1) = T(u_1)$

$T/W(u_2) = T(u_2)$

$$T/W(u_1+u_2) = T(u_1+u_2)$$
$$= T(u_1) + T(u_2)$$
$$= T/W(u_1) + T/W(u_2)$$

$$T/W(\alpha u_1) = T(\alpha u_1)$$
$$= \alpha T(u_1)$$
$$= \alpha\, T/W(u_1)$$

$\therefore$ T/W is linear.

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